

Kyaw (Kelvin) Ha

Nutley, NJ | kelvinthiha@rutgers.edu | [linkedin.com/in/kyawha](https://www.linkedin.com/in/kyawha) | kelvinthiha17.github.io/portfolio

EDUCATION

Rutgers University, New Brunswick, NJ

Expected May 2027

Bachelor of Science, Computer Engineering

Ohlone College, Fremont, CA

May 2024

Associate of Science, Computer Engineering

TECHNICAL SKILLS

Languages: Verilog, SystemVerilog, Embedded C, C++, Java, Python, MATLAB, HTML, CSS, JavaScript

Embedded / Firmware: STM32CubeIDE, STM32CubeMX, HAL Drivers, FreeRTOS (CMSIS-RTOS2), UART, PWM, Timer Input Capture, Interrupt-Driven Programming

Digital Design / Tools: FPGA Design (Vivado), Icarus Verilog, GTKWave, QSPICE, LTSpice, PCB Design, Schematic Capture, Soldering, Logic Analyzers, Oscilloscopes

Core Knowledge: RTL Design & Verification, CPU Microarchitecture & Pipelining, ISA Design, Hazard Detection & Forwarding, ASIC Design Flow, Digital Logic Design, Embedded Systems, Circuit Design, Technical Documentation

PROJECTS

MiniRISC16 Pipelined Processor

Verilog, RTL Design, CPU Microarchitecture

- Designed and implemented a 16-bit RISC CPU in Verilog supporting R-type ALU operations (ADD, SUB, AND, OR, XOR), load/store memory access (LW, SW), and conditional branching (BEQ).
- Extended the single-cycle architecture to a 5-stage pipelined datapath (IF/ID/EX/MEM/WB) with data hazard detection, operand forwarding, load-use stalling, and branch flushing.
- Verified correct pipeline behavior through comprehensive simulation testbenches using Icarus Verilog and GTKWave, confirming hazard resolution and instruction execution across all implemented opcodes.

STM32 Embedded Sensor & Control System

STM32CubeIDE, Embedded C, FreeRTOS

- Developed multi-peripheral embedded firmware on an STM32 (Cortex-M0+) microcontroller, integrating GPIO-timed sensor interfacing, hardware PWM generation for servo motor control, and interrupt-driven timer input capture for precision pulse-width signal timing.
- Implemented a FreeRTOS-based multitasking architecture (CMSIS-RTOS2) with independently scheduled concurrent tasks using non-blocking delays, alongside real-time UART telemetry over a serial debug console.
- Diagnosed and resolved HAL driver state-machine and peripheral routing conflicts through systematic register-level debugging, restoring reliable UART communication and timer-interrupt behavior.

Traffic Signal State Machine

Verilog, TTL Logic, QSPICE, LTSpice

- Designed and simulated the full circuit schematic in QSPICE and LTSpice prior to hardware assembly, validating logic behavior and timing before physical implementation.
- Built and assembled the circuit on a breadboard using TTL logic ICs, implementing Red → Green → Yellow state transitions through combinational and sequential gate logic with custom clock generation circuitry.
- Performed functional verification of all state transitions against a working traffic light setup, confirming correct logic behavior across every operating phase using oscilloscopes and logic analyzers.

Huffman Coding Compressor

C++

- Built a lossless file compression and decompression tool using Huffman Trees and priority queues; achieved ~40% size reduction on large text datasets.
- Implemented dynamic encoding for any user-input text file via CLI with full encode/decode round-trip verification, demonstrating strong low-level C++ and algorithmic problem solving.

Personal Portfolio Website

HTML, CSS, JavaScript

- Responsive site with AOS scroll animations, dark-mode toggle, and project detail pages; hosted on GitHub Pages at kelvinthiha17.github.io/portfolio.

EXPERIENCE

R&B Auto Works, Yangon, Myanmar

Jan 2019 – Jan 2020

Technical Intern

- Assisted in automotive diagnostics and hands-on component-level repair, applying systematic troubleshooting to identify and resolve failures.
- Tracked inventory, repair data, and supply usage in Microsoft Excel; documented repair processes and maintained technical accuracy across all maintenance records.

Happy Lemon, Newark, CA

Jan 2023 – Sep 2023

Barista / Team Member

- Managed high-volume orders in a fast-paced environment while maintaining accuracy and customer satisfaction.
- Trained and onboarded new employees on drink preparation, contributing to improved team consistency.